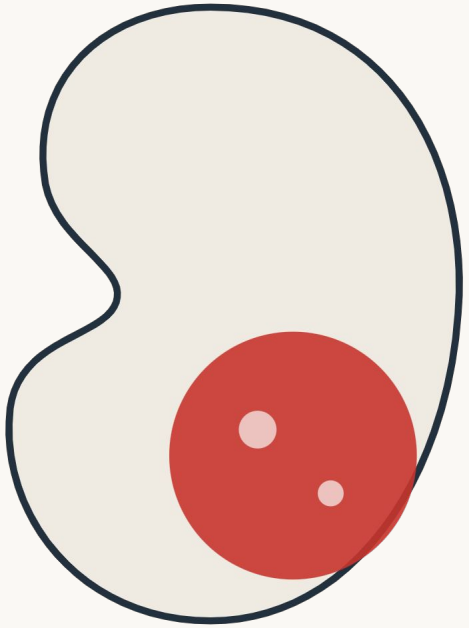
Wilms tumour



TARGET-WT dataset
from Tumor classification to syndromic cases

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The most common kidney cancer of childhood

Also called nephroblastoma, it is an embryonal cancer that grows from kidney tissue which never finished maturing.

>80%*
of childhood kidney tumours

2–5 y*
usual age at diagnosis

5–9%*
in both kidneys

~90%*
survive today

* Spreafico F, Fernandez CV, Brok J, et al. Wilms tumour. Nat Rev Dis Primers 2021;7:75.

Embryonic kidney that keeps growing

Nephron progenitors → Nephrogenic rests → Wilms tumour

Normally mature into nephrons before birth. Marked by SIX2 and CITED1.

Islands of embryonic tissue left behind, the precursor lesions.

Further genetic hits, and the growth is no longer controlled.

Under the microscope the classic tumour is *triphasic*, a mix of blastema, epithelium and stroma.

Recurrent genes include WT1, CTNNB1, AMER1, IGF2 at 11p15, SIX1, SIX2, DROSHA and DICER1. TP53 goes with anaplasia.

Anaplasia splits the patients in two

92%*

Favorable histology

No anaplasia. The classic triphasic tumour, very sensitive to chemotherapy, with an excellent outcome at most stages.

7–8%*

Unfavorable, anaplastic

Very large, dark nuclei and abnormal multipolar mitoses, either focal or diffuse. Often TP53-mutant* and resistant to chemotherapy.

Unfavorable is about chemoresistance more than speed. These tumours survive the treatment that cures the others.

* Spreafico F, Fernandez CV, Brok J, et al. Wilms tumour. Nat Rev Dis Primers 2021;7:75.

Beckwith–Wiedemann syndrome

An imprinting disorder of 11p15.5 (IGF2, H19, CDKN1C) seen in about 1 in 10,000 births**. The same region is often altered in sporadic Wilms tumour.

Children can show macroglossia, overgrowth or lateralised overgrowth, omphalocele, neonatal hypoglycaemia and ear creases or pits.

Higher-risk subtypes are followed with an abdominal ultrasound every 3 months until about age 7**.

Molecular cause	Share*	Tumour risk*
IC2 loss of methylation	52%	2.6%, Wilms 0.2%
Paternal UPD 11p15	19%	16%, Wilms 7.9%
IC1 gain of methylation	8%	28%, Wilms 24%
CDKN1C mutation	4%	6.9%, neuroblastoma

Overall childhood tumour risk is about 8%*. Wilms tumour in BWS tends to come earlier and to be bilateral more often. WAGR and Denys–Drash, both WT1 syndromes, also predispose.

* Maas SM, Vansenne F, Kadouch DJM, et al. Phenotype, cancer risk, and surveillance in Beckwith–Wiedemann syndrome depending on molecular genetic subgroups. Am J Med Genet A 2016;170:2248–2260.

** Brioude F, Kalish JM, Mussa A, et al. Clinical and molecular diagnosis, screening and management of Beckwith–Wiedemann syndrome: an international consensus statement. Nat Rev Endocrinol 2018;14:229–249.